

Exercise 8 – Number System (Scientific Notation)

Scientific number

$$\Rightarrow (\text{Fraction}) \times (\text{Base})^{\text{Exponent}}$$

$$\Rightarrow \text{decimal: } 1.2 \times 10^4$$

$$\Rightarrow \text{binary: } 1.01 \times 2^2$$

Example:

$$\begin{aligned} (245.34)_{10} &\Rightarrow 245.34 \\ &\Rightarrow 2.4534 \times 10^2 \\ \text{Fraction: } 2.4534 \quad &\text{Exponent} = 2 \end{aligned}$$

$$\begin{aligned} (0.000245)_{10} &\Rightarrow 0.000245 \\ &\Rightarrow 2.4534 \times 10^{-4} \\ \text{Fraction: } 2.4534 \quad &\text{Exponent} = -4 \end{aligned}$$

$$\begin{aligned} (1101.1111)_2 &\Rightarrow 1101.1111 \\ &\Rightarrow 1.1011111 \times 2^3 \\ \text{Fraction: } (1.1011111)_2 \quad &\text{Exponent} = 101_2 \end{aligned}$$

$$\begin{aligned} (0.001011111)_2 &\Rightarrow 0.001011111 \\ &\Rightarrow 1.011111 \times 2^{-3} \\ \text{Fraction: } (1.1011111)_2 \quad &\text{Exponent} = (-3)_{10} \Rightarrow (111)_2 \quad [2\text{'s compliment}] \end{aligned}$$

Question: Write following numbers in scientific form:

- (a) 145.6 (b) $(111011.111)_2$ (c) $(0.0035.21)_8$ (d) $(0.00000111)_2$

Question: Convert form scientific form to numbers

- (a) 2.345×10^{-2} (b) $(1.10101110 \times 2^6)_2$ (c) $(1.234567 \times 8^3)_8$ (d) $(0.11101 \times 2^{-3})_2$